

Department of Statistics
University of Rajshahi
B.Sc. (Hons.) Part – IV Practical Assignment, May 29, 2026
Subjects: Statistics, Course: B.Stat. 401
Course Title: Multivariate Analysis

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Q0. Data Characteristics.

(a). Write the names and types of all variables given in the generated data frame named “data”.

Answer:

Variable Name	Type
GDP growth	Numeric
Unemployment	Numeric
Inflation	Numeric
Income per capita	Numeric
Crime rate	Numeric
Population density	Numeric
Social support	Numeric
Life expectancy	Numeric
Infant mortality	Numeric
Health expenditure	Numeric
Literacy rate	Numeric
School enrollment	Numeric
Teacher student ratio	Numeric
Air quality index	Numeric
Green space	Numeric
Water quality	Numeric

(b). What is the purpose of the command scale(data)?

Answer:

In data analysis and statistics, the scale(data) command is commonly used to standardize data. It is widely known from R, although similar methods exist in other programming languages, such as StandardScaler in Python. The main purpose of standardization is to place variables on a comparable scale.

When scale(data) is applied to a numeric column, two main operations are performed:

Centering (Subtracting the Mean):

First, the average (mean) of the column is calculated and subtracted from each observation. This shifts the data so that the transformed variable has a mean of zero.

Scaling (Dividing by the Standard Deviation):

Next, each centered value is divided by the standard deviation of the column. This rescales the data so that the transformed variable has a standard deviation of one (and variance equal to one).

As a result, each observation is converted into a standardized score (Z-score), calculated as:

$$z = (x - \mu) / \sigma$$

where μ represents the mean and σ represents the standard deviation.

(C). Interpret the output of the command summary(data_scaled).

Answer:

Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max
GDP_growth	-2.775	-0.7135	-0.0337	0	0.7254	3.0747
unemployment	-2.5551	-0.6615	-0.0427	0	0.7028	3.0328
inflation	-1.4142	-0.8077	-0.1358	0	0.5771	3.604
income_per_capita	-2.7165	-0.7153	-0.0302	0	0.6802	2.9208
crime_rate	-2.2267	-0.6601	-0.0704	0	0.7101	3.2536
population_density	-2.8258	-0.6378	-0.0442	0	0.6704	2.7324
social_support	-2.5406	-0.6132	-0.0841	0	0.6725	2.7832
life_expectancy	-2.6902	-0.7182	0.01087	0	0.6571	3.2081
infant_mortality	-2.6581	-0.7189	-0.0191	0	0.666	2.9335
health_expenditure	-3.0835	-0.668	0.04436	0	0.6929	2.6438
literacy_rate	-3.2522	-0.7139	-0.0104	0	0.7462	2.9078
school_enrollment	-2.9799	-0.7016	0.00955	0	0.633	2.6802
teacher_student_ratio	-2.5844	-0.7131	-0.0138	0	0.7041	2.8968
air_quality_index	-1.4655	-0.7809	-0.1807	0	0.6437	3.4999
green_space	-2.8718	-0.6587	0.03274	0	0.6211	3.0674
water_quality	-3.4762	-0.6514	0.0752	0	0.6492	3.0197

Since the data were standardized using scale(data), several important observations can be made:

1. Mean values are approximately zero for all variables, confirming that the centering process was successfully performed.
2. The median values are generally close to zero, indicating that most variables are reasonably centered and show relatively balanced distributions around the mean.

3. The minimum and maximum values represent the range of standardized observations, expressed in terms of standard deviation units from the mean. These values show how far observations deviate from the center after scaling.
4. The first and third quartiles describe the spread of the middle 50% of observations following standardization, providing insight into the variability of each variable.
5. Variables with relatively large absolute minimum or maximum values may indicate the presence of extreme observations or potential outliers, as these values lie further away from the mean compared to most observations in the dataset.

Q1. PRINCIPAL COMPONENT ANALYSIS (PCA).

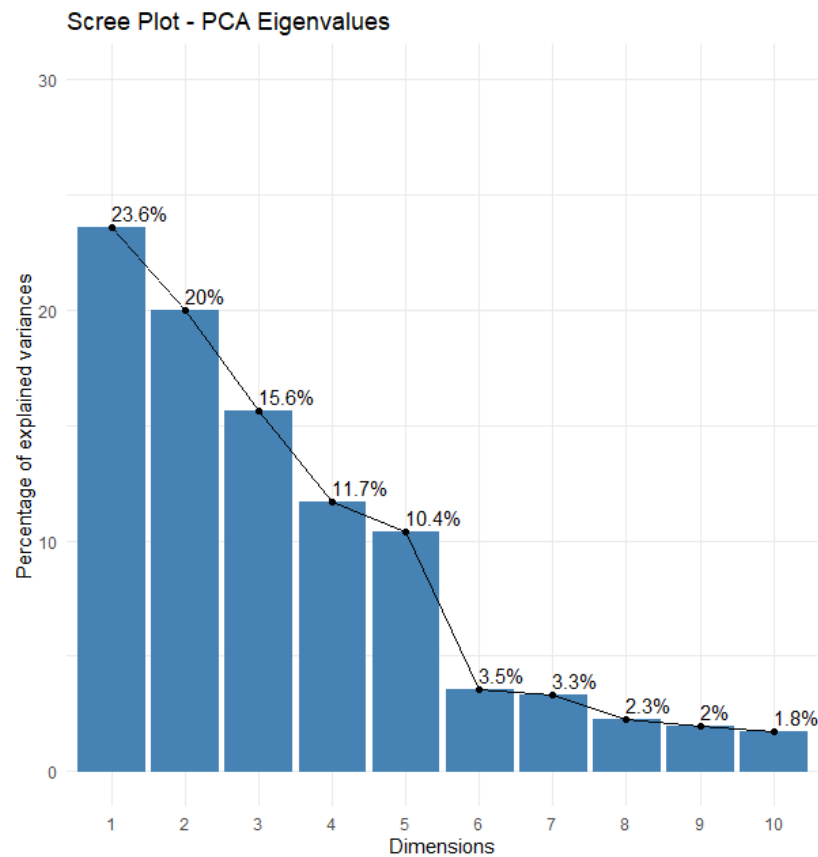
(a). Find the eigenvalues of the covariance matrix for the data frame data_scaled.

Answer:

Component	Eigenvalue
PC1	3.769
PC2	3.1999
PC3	2.5038
PC4	1.8701
PC5	1.6609
PC6	0.5678
PC7	0.5289
PC8	0.3602
PC9	0.3172
PC10	0.2809
PC11	0.2567
PC12	0.2372
PC13	0.172
PC14	0.1142
PC15	0.0926
PC16	0.0684

(b). Perform PCA. Draw a scree plot and interpret it.

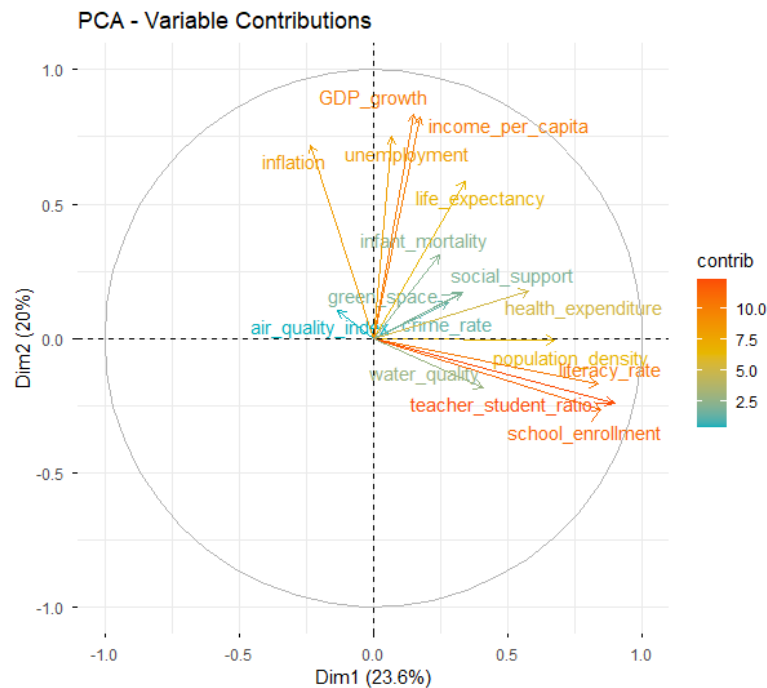
Answer:



The scree plot indicates that the first five principal components are important because they explain about 81.3% of the total variance. PC1 explains the maximum variance, followed by PC2 and PC3. After PC5, the curve becomes flatter, meaning the remaining components explain very small amounts of variance and can be considered less important. Therefore, retaining the first five components is suitable for summarizing most of the information in the dataset.

(c). Visualize variable contributions and interpret the graph.

Answer:



The PCA biplot indicates that Dim1 explains 23.6% and Dim2 explains 20% of the total variance, together explaining 43.6%. Variables like literacy rate, school enrollment, population density, and teacher-student ratio contribute strongly to Dim1, suggesting that Dim1 represents education and social development. Variables such as GDP growth, income per capita, inflation, unemployment, and life expectancy contribute more to Dim2, indicating an economic and wellbeing dimension. Variables pointing in the same direction are positively correlated, while variables in opposite directions are negatively correlated.

(d). Compute the “Explained variance by each component” and interpret it.

Answer:

Component	Variance Explained
PC1	23.557
PC2	19.999
PC3	15.649
PC4	11.688
PC5	10.381
PC6	3.549
PC7	3.306
PC8	2.252
PC9	1.982
PC10	1.756
PC11	1.604
PC12	1.483

PC13	1.075
PC14	0.714
PC15	0.579
PC16	0.427

The PCA result shows that the first principal component explains the highest variance, 23.557%, followed by PC2 with 19.999%. This means PC1 contains the most important information from the original variables. The first five components together explain 81.274% of the total variance, so they retain most of the information in the dataset. Components after PC5 explain very small amounts of variance and may be considered less important for interpretation.

Q2. FACTOR ANALYSIS (FA)

(a). Apply the Kaiser-Meyer-Olkin (KMO) test and the Bartlett's test of sphericity. Comment on these two tests.

Answer:

- Kaiser–Meyer–Olkin (KMO) Measure of Sampling Adequacy

Overall KMO (MSA) = 0.76

Variable	MSA
GDP Growth	0.81
Unemployment	0.77
Inflation	0.9
Income per Capita	0.83
Crime Rate	0.67
Population Density	0.89
Social Support	0.61
Life Expectancy	0.69
Infant Mortality	0.6
Health Expenditure	0.88
Literacy Rate	0.8
School Enrollment	0.85
Teacher–Student Ratio	0.79
Air Quality Index	0.79
Green Space	0.68
Water Quality	0.66

- Bartlett's Test of Sphericity

Statistic	Value
Chi-Square (χ^2)	3482.133
Degrees of Freedom (df)	120
p-value	< 0.001

Comment: The Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy yielded an overall value of **0.76**, indicating that the data are suitable for factor analysis. According to

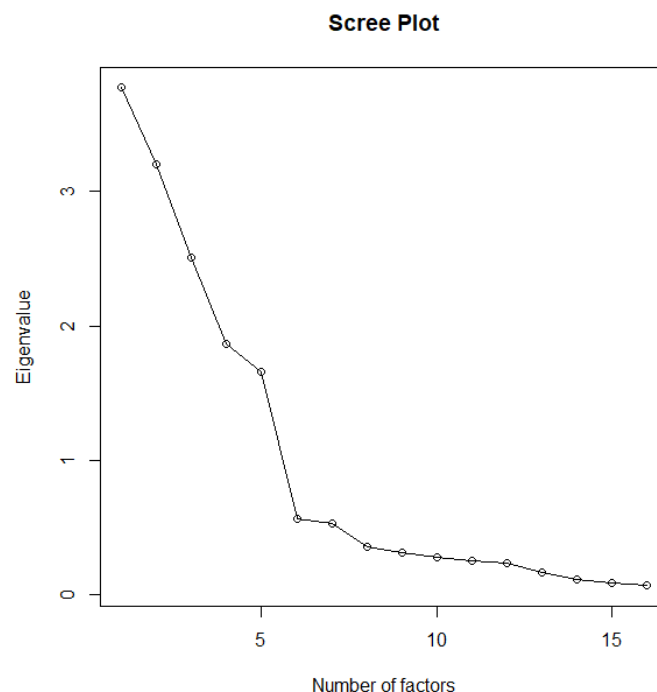
commonly accepted guidelines, a KMO value above 0.70 suggests that the variables share sufficient common variance for reliable factor extraction. Furthermore, all individual MSA values exceeded the minimum acceptable threshold of 0.50, indicating that each variable contributes adequately to the factor structure.

Bartlett's Test of Sphericity was highly significant ($\chi^2 = 3482.133$, $df = 120$, $p < 0.001$), rejecting the null hypothesis that the correlation matrix is an identity matrix. This result confirms the presence of significant correlations among the variables and demonstrates that the data possess an underlying structure suitable for factor analysis.

Taken together, the KMO and Bartlett's test results indicate that the dataset satisfies the key assumptions for factor analysis and that the extracted factors are likely to be meaningful and interpretable.

(b). Select the suitable number of factors based on the scree plot and assign that value to Nfacs.

Answer:



The scree plot shows a clear elbow at the 5th factor, after which the eigenvalues decrease very slowly. Therefore, the first five factors explain most of the variation in the data. Hence, the suitable number of factors is $N_{facs} = 5$.

(c). Perform FA and interpret the results. Provide possible names for the assumed factors.

Answer:

H₀: The five-factor model provides a good fit to the data.

V/S

H₁: H₀ is not true

Rotated Factor Loading Matrix (Varimax Rotation)

Variable	Education Factor (F1)	Economic Factor (F2)	Health Factor (F3)	Social Factor (F4)	Environmental Factor (F5)
Literacy Rate	0.953				
Teacher–Student Ratio	0.936				
School Enrollment	0.927				
Unemployment		0.865			
GDP Growth		0.842			
Income per Capita		0.829			
Inflation		0.571			
Infant Mortality			0.972		
Life Expectancy			0.853		
Health Expenditure			0.705		
Social Support				0.926	
Crime Rate				0.859	
Population Density				0.591	
Water Quality					0.888
Green Space					0.75
Air Quality Index					–0.591

Extraction Method: Maximum Likelihood (ML)

Rotation Method: Varimax Rotation

Number of Factors Retained: 5

Variance Explained by the Extracted Factors

Factor	Proposed Name	Variance Explained (%)	Cumulative Variance (%)
F1	Education Factor	19.5	19.5
F2	Economic Factor	16.5	36
F3	Health Factor	14.2	50.3
F4	Social Factor	12.6	62.9
F5	Environmental Factor	11.0	73.9

Factor analysis was performed using the Maximum Likelihood (ML) extraction method with Varimax rotation. Based on the Kaiser criterion (eigenvalues greater than one), five factors

were retained. The results show that the five-factor solution explains approximately 73.9% of the total variance in the dataset. The model fit is highly satisfactory, as indicated by the likelihood ratio test ($\chi^2 = 35.72$, $df = 50$, $p = 0.936$), suggesting that the five-factor model adequately reproduces the observed correlation structure. In addition, the RMSEA value of 0.000 and RMSR value of 0.011 further confirm the excellent fit of the model.

Based on the rotated factor loadings, Factor 1 has high loadings on literacy_rate, teacher_student_ratio, and school_enrollment and may therefore be interpreted as an Education Factor. Factor 2 loads highly on GDP_growth, unemployment, income_per_capita, and inflation, representing an Economic Factor. Factor 3 is characterized by strong loadings on infant_mortality, life_expectancy, and health_expenditure, and can be named the Health Factor. Factor 4 has high loadings on social_support, crime_rate, and population_density, reflecting a Social Factor. Finally, Factor 5 loads strongly on water_quality, green_space, and air_quality_index, indicating an Environmental Factor.

Therefore, the factor analysis successfully reduces the original sixteen variables into five meaningful latent factors: Education, Economic, Health, Social, and Environmental. These factors capture the major dimensions underlying the dataset and provide a more concise representation of the observed variables.

Q3. CANONICAL CORRELATION ANALYSIS (CCA)

(a). Interpret the CCA results provided by the given R code.

Answer:

Canonical Correlations

Canonical Function	Canonical Correlation
1	0.654
2	0.6137
3	0.3923
4	0.2148
5	0.1725
6	0.0999

The first two canonical correlations (**0.654** and **0.614**) indicate moderately strong relationships between the two variable sets. The third canonical correlation (**0.392**) indicates a moderate association, whereas the remaining correlations are relatively weak.

Significance of Canonical Functions

Function Tested	Wilks' Lambda	F-value	p-value	Interpretation
1–6	0.2766	8.776	<0.001	Significant
2–6	0.4833	6.522	<0.001	Significant
3–6	0.7752	3.173	<0.001	Significant
4–6	0.9162	1.713	0.0437	Significant
5–6	0.9606	1.473	0.164	Not Significant
6	0.99	0.978	0.403	Not Significant

The Wilks' Lambda tests indicate that the first four canonical functions are statistically significant, while the fifth and sixth functions are not significant. Therefore, interpretation should focus primarily on the first four canonical dimensions, particularly the first two due to their stronger canonical correlations.

Canonical Correlation Analysis (CCA) was performed to examine the relationship between two sets of variables: the Economic–Social variables (Set X) and the Health–Education–Environment variables (Set Y).

The analysis produced six canonical correlations: 0.654, 0.614, 0.392, 0.215, 0.173, and 0.100. The first two canonical correlations indicate moderately strong relationships between the two variable sets, while the remaining correlations are comparatively weaker.

The significance of the canonical functions was assessed using Wilks' Lambda. The first four canonical functions were found to be statistically significant ($p < 0.05$), whereas the fifth and sixth functions were not significant ($p > 0.05$). Therefore, only the first four canonical functions are considered meaningful for interpretation.

The first canonical function ($r = 0.654$) is mainly influenced by GDP growth, unemployment, income per capita, and population density from Set X, and by life expectancy, infant mortality, and literacy rate from Set Y. This function suggests a strong association between economic conditions and human development outcomes. Better economic performance is associated with higher life expectancy, lower infant mortality, and improved literacy levels.

The second canonical function ($r = 0.614$) is primarily characterized by population density, inflation, and crime rate from Set X and school enrollment from Set Y. This indicates a relationship between demographic and economic conditions and educational outcomes.

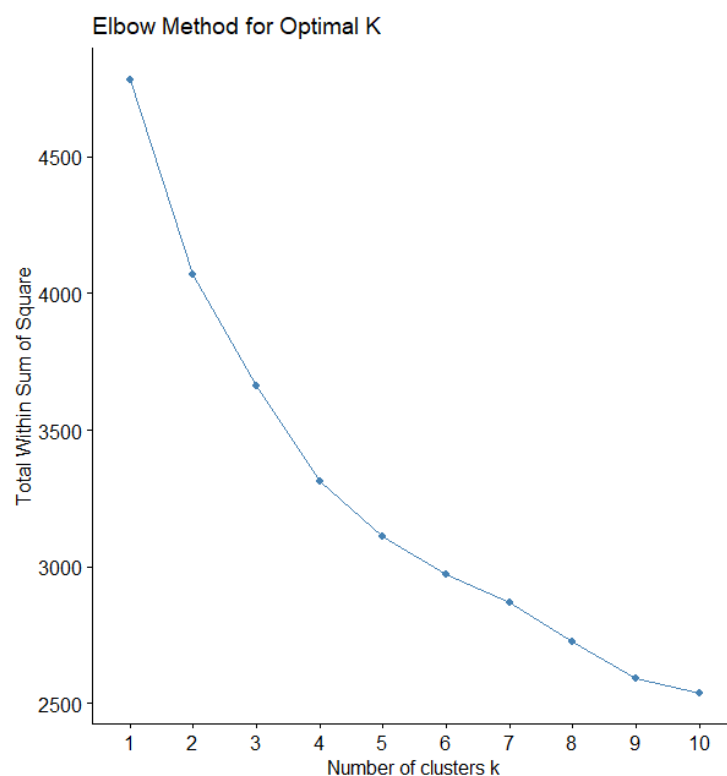
The third canonical function ($r = 0.392$) is mainly driven by unemployment, GDP growth, and crime rate from Set X and infant mortality from Set Y, suggesting that labor market and economic conditions are related to health outcomes.

Overall, the CCA results reveal significant relationships between the Economic–Social and Health–Education–Environment variable sets. The first two canonical functions explain the strongest associations, indicating that socioeconomic conditions play an important role in determining health, education, and environmental outcomes. Therefore, improvements in economic and social factors are likely to contribute positively to overall human development.

Q4. K-MEANS CLUSTER ANALYSIS

(a). Determine the optimal number of clusters using the Elbow method and Silhouette method, and assign this value to K.value.

Answer:



The optimal number of clusters was determined using both the Elbow Method and the Silhouette Method. The Elbow Plot indicated that the rate of decrease in slowed noticeably after $K = 5$, suggesting four clusters as a reasonable choice.

Method	Optimal Number of Clusters
Silhouette Method	2

Although the elbow method suggests $K = 5$, the silhouette method suggests $K = 2$. Since the silhouette method directly measures cluster separation and cohesion, $K = 2$ can be considered

the more statistically supported choice. However, if the research objective needs more detailed grouping, $K = 5$ may also be considered.

(b). Perform the K-means cluster analysis with the assigned K.value and interpret briefly the outputs provided by the given R code.

Answer:

Cluster Sizes:

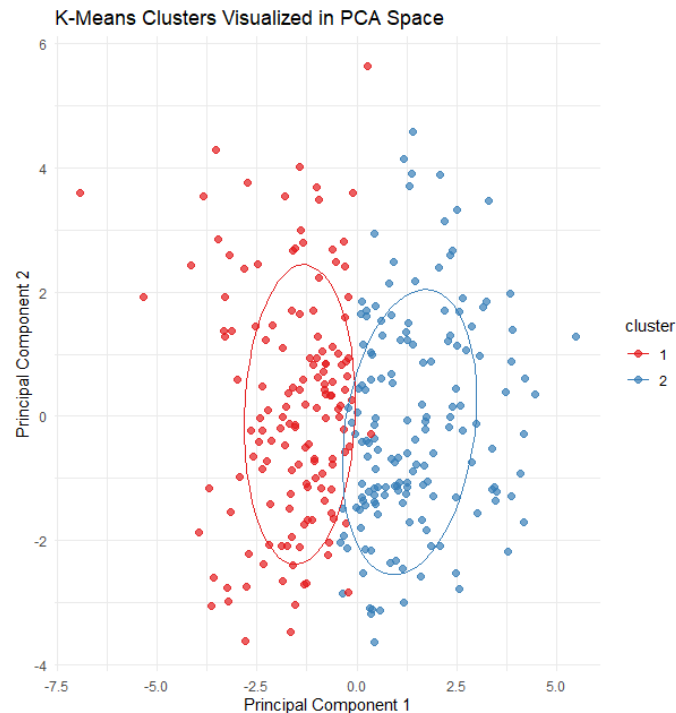
Cluster	Number of Observations	Percentage (%)
Cluster 1	143	47.7
Cluster 2	157	52.3
Total	300	100

The silhouette method identified $K = 2$ as the optimal number of clusters. Consequently, K-means clustering was performed with two clusters. The analysis classified the 300 observations into Cluster 1 (143 observations, 47.7%) and Cluster 2 (157 observations, 52.3%).

Key Characteristics of Clusters (Standardized Centers):

Variable	Cluster 1	Cluster 2
GDP Growth	-0.094	0.086
Unemployment	-0.045	0.041
Income per Capita	-0.115	0.105
Life Expectancy	-0.208	0.189
Health Expenditure	-0.442	0.403
Literacy Rate	-0.787	0.717
School Enrollment	-0.768	0.699
Teacher–Student Ratio	-0.805	0.733
Green Space	-0.219	0.199
Water Quality	-0.311	0.283

Examination of the cluster centers reveals clear differences between the two groups. Cluster 2 exhibits above-average values for important development indicators such as income per capita, life expectancy, health expenditure, literacy rate, school enrollment, teacher–student ratio, green space, and water quality, while Cluster 1 shows below-average values for these variables. Therefore, Cluster 2 may be interpreted as a High Development Cluster, whereas Cluster 1 may be interpreted as a Low Development Cluster.



The K-means cluster plot shows the observations divided into two groups in PCA space.

Cluster 1 is mainly located on the left side of Principal Component 1, while Cluster 2 is mainly located on the right side. This means the two clusters are mostly separated by PC1.

There is some overlap between the two clusters near the center, so the separation is moderate, not perfect. However, the two ellipses show two clear general groupings in the data.

Declaration Statement:

“I, myself, have written the answers and have not copied text from the solutions of any classmates. I have not used AI text generators or AI-powered paraphrasing tools to create any part of the content.”

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